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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

```
# Load numerical dataset
data = pd.DataFrame({
    'Feature1': [2.5, 0.5, 2.2, 1.9, 3.1],
    'Feature2': [2.4, 0.7, 2.9, 2.2, 3.0],
    'Feature3': [1.1, 0.3, 1.5, 1.3, 1.9]
})
print("Original Dataset:");
print(data);
```

Original Dataset:

	Feature1	Feature2	Feature3
0	2.5	2.4	1.1
1	0.5	0.7	0.3
2	2.2	2.9	1.5
3	1.9	2.2	1.3
4	3.1	3.0	1.9

```
#1. Standardization
scaler = StandardScaler()
scaled_data = scaler.fit_transform(data)

print("Standardized Dataset:");
print(scaled_data);
```

Standardized Dataset:

[	0.53102066	0.193687	-0.22613351]
[	-1.77776482	-1.86423738	-1.73369023]
[	0.18470284	0.79895888	0.52764485]
[	-0.16161498	-0.04842175	0.15075567]
[	1.22365631	0.92001325	1.28142321]

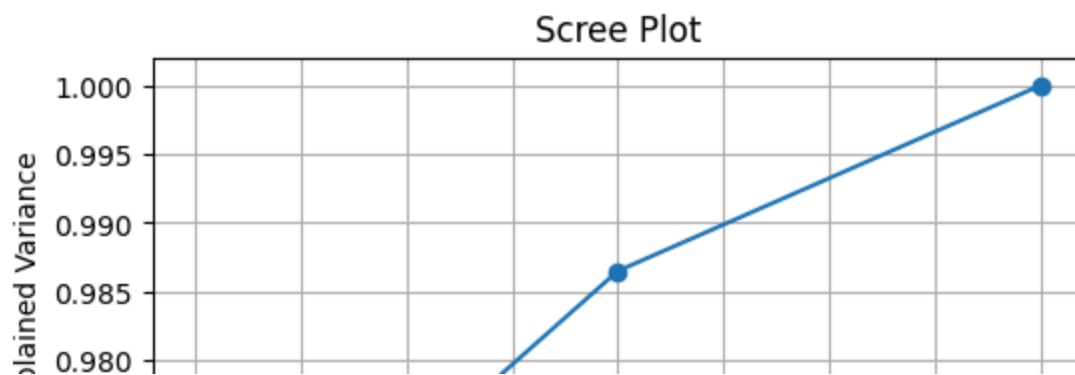
```
#2. PCA
pca = PCA()
pca_data = pca.fit_transform(scaled_data)

#3. Scree Plot
explained_variance = pca.explained_variance_ratio_

print("\nExplained Variance Ratio:");
print(explained_variance);
```

```
Explained Variance Ratio:
[0.95925771 0.02723948 0.01350281]
```

```
plt.figure(figsize=(6,4))
plt.plot(np.cumsum(explained_variance), marker='o')
plt.xlabel('Number of Principal Components')
plt.ylabel('Cumulative Explained Variance')
plt.title('Scree Plot')
plt.grid()
plt.show()
```



```
#4. Dimensionality Reduction to 2D
pca_2 = PCA(n_components=2)
reduced_data = pca_2.fit_transform(scaled_data)
```

```
#5. 2D Visualization
plt.figure(figsize=(6,5))
plt.scatter(reduced_data[:, 0], reduced_data[:, 1])
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
```

```
plt.title('PCA - 2D Data Visualization')  
plt.grid()  
plt.show()
```

